Promise-Physics

A diagnostic for capital asset acquisition - v1.0

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A promise hangs from measure, not will.

Abstract

For federal acquisition, fixed price is becoming the default. Against a scope with no native unit, a fixed price produces the outcome everyone fears: the exchange between scope and dollars runs until the calendar is exhausted, and whatever is on the table at that moment becomes the project. The missing piece is not a contract type. What's missing is a measured and ranked *ask*.

The Department of Energy built a measured and ranked ask. The Research Support Facility at NREL was procured at a fixed price (set before the performance goals existed), against a scope ranked in three tiers, with a draft RFP put in front of the final competitors before competition opened. The owner's stated reason for being able to use a firm fixed-price contract was that the contractors knew exactly what was wanted, which reduced their risk. The record is public and cited at the end of this paper.

This paper states the *physics* under that result. Three laws — **Sovereignty**, **Precedence**, **Tolerance** — and the six elements the laws produce. It closes with one compound question that reads any solicitation, and returns findings with addresses rather than a score.

Nothing here is a method. Nothing here can be obeyed or violated. Each statement is made so that a project could prove the promise-physics wrong.

The axiom

Between two, a promise hangs from common measures, not individual will.

The promise holds because it can be read, and read the same way by both parties. *Between* is the operative word: a measure on only one party's side of the table is not between anything. It is an assertion, and a promise made against an assertion carries two readings from two parties from the outset.

The primary breakdown in keeping a promise is not bad faith. It is two honest parties standing on different measures and discovering it after a handshake.

A solicitation is a written promise. What follows reads it as one.

The three laws

I · Sovereignty — conservation of accountability

A promise binds only where each party made it.

Divided authorship does not divide the promise. It strips the accountable address off it, and unaddressed accountability defaults to the party that cannot walk away from the asset.

Predicts: divide authority over a decision and its cost does not disappear. It becomes unassignable, and it lands on the owner.

II · Precedence — understanding before promise

A promise binds only to the extent it was made.

The completeness of the ask falls to the owner. The comprehension of the ask falls to the contractor. Neither borrows from the other's failure. Whatever the contractor volunteers is asked: a claim made is a claim enforceable, at the same grade as a requirement.

Understanding is free only while refusal is free. After commitment, every unresolved question is priced by a party facing no competition.

Predicts: whatever was left unpromised at award is where the change orders come from. The list is predictable in advance.

III · Tolerance — limit of restraint

A promise holds only to the limit both parties can bear to stay in their own domain.

Restraint is the only proof of that limit, and it is paid continuously: every crossing declined, every invitation refused. Past the limit, the parties fall back on will — which is what the measure was there to replace.

Predicts: Laws I and II operate at full strength only up to this limit. Above it, the contract describes something that will not happen.

The laws produce a boundary, and the boundary is where authority and liability live. Before award the owner governs: what the finished thing must do, ranked and measured. After award the contractor governs: means, methods, design. The owner's remaining work is confirming and verifying against what was promised (published). The owner may still change the problem — but that is a new promise, understood and priced before it binds, not an amendment.

The three objects

For capital assets, every promise has three dimensions: **scope, dollars, time.** Close all three and a promise exists. Leave one or more open and there is no promise — only an intention with a hole in it, and the hole is where a counterparty or a tribunal takes jurisdiction.

The three arrive already fixed. **The appropriation fixed the money, the period of availability fixed the time, and the authorized purpose fixed the scope.** The *owner* set none of them. They restate what the appropriation already fixed, in terms a market can read and respond. These are the only true constraints — conditions of law, not curable by writing.

The three are not alike. **Dollars** are a stock: self-measuring, holdable, arriving from outside before the project exists. The owner's question is never *what does this outcome cost*; it is *what outcome can I get for the number I was given.* **Time** is a flow: it cannot be held, and it is spent in absence rather than in events, so it produces no moment at which anyone can refuse. **Scope** is the only dimension without a native unit. It cannot be read until it is given one, and ambiguity in scope is not a weak promise — it is the absence of a promise.

Time is consumed in proportion to the ambiguity remaining in the other two. Every scope question is a dollars question, every dollars answer reopens scope, and the exchange may end when the calendar is exhausted, not when the answer arrives.

The six elements

What a written promise is made of. Each closes one exposure. A void is not neutral; it is an exposure carried by someone, usually the party who cannot walk away from the asset.

	Element	The exposure it closes	Source
1	**Fixed price**	The owner cannot be outbid past their money; the contractor is not bidding against themselves	The Object
2	**Fixed date**	Neither party carries an unpriced calendar	The Object
3	**Ranked scope, flattened at award**	The owner can be told no without failing; the contractor can decline a tier without breaching; no tier survives as an excuse	The Object
4	**Common measure and its discharge**	Neither party is held to the other's reading; the contractor's reach is finite and releasable	Axiom · Precedence
5	**Costless no**	Neither party is bound beyond what they individually affirmed	Precedence
6	**Exclusive domains**	No joint decision, therefore no unaddressed obligation	Sovereignty

The elements are not independent. Remove one and its exposure returns, and the party holding it stops reaching on that item at once.

Element 4 requires the procedure, the conditions, the instrument, and the threshold — stated, never implied — together with who reads, the owner's window, and the sole ground for withholding. A measure exists only to be discharged: the read is a fact, the release is an act. A measure attached to an unknowable release is a promise the contractor cannot reach, and it prices exactly like any open and uncertain dimension.

Element 5 binds on an affirmative act, not an internal state. A contractor who later reports they did not understand a line is asserting something nobody can check; a contractor who affirmed the line, on the record, has spent their own comprehension into the affirmation. This is why the reading runs line by line: every line draws either an affirmation or a defect report, and there is no third state.

The diagnostic

One purposeful compound question, asked of the solicitation:

Is the RFP clear, and if so, can the *required* scope be achieved at the price (IGE) and time stated?

The *and if so* is load-bearing. The second dependent question requires the *yes* as an answer to the preposition. A solicitation that fails the first half can not get to a second answer. An achievability reading taken against an unclear ask is not a weak reading; the answer is baseless and can't be known.

Is it clear reads elements 3, 4 and 6 — line by line, in the document.

Can it be achieved at the price and time stated reads elements 1, 2 and 5.

Unclear

Always curable, and the cure is always the same thing. A line that reads two ways is a line with no common measure on it — a missing procedure, condition, instrument, or threshold, or a release nobody can know. Not a communication problem. A defect with a known repair, fixed at the desk, in the owner's domain, before commitment, in relative short order, at no cost to anyone.

Unachievable

Not curable by writing. Nothing is wrong with the ask. The finding is that the number and the required scope do not converge in this market, in this window.

The number is the IGE — the owner's own estimate, produced by the owner, held by the owner, never read by the other party. **An IGE is an assertion on one party's side of the table.** Its congruence with the market is unknown to everyone, including the owner, until the market reads it against the ask.

An unreachable *required* tier is therefore a report about the appropriation, not a decision available to the owner. The *required* tier is the authorized purpose restated; cutting into it would spend the money on something other than what it was authorized for. The queue absorbs the market's report **above** the floor, where the owner's own reach lives. At the floor, the report goes back to the party that set the three objects.

When the reading is worth taking

Market research is required before soliciting and remains available up to the point competition begins. The regulation put the authority precisely where the instrument works, and the physics says why.

Before competition opens, a party can answer *not achievable* and lose nothing. The no is free, which is what makes the yes evidence. **After selection,** there is nothing left to decline, and every affirmation given then is compliance. The activity is identical; only the audience has become more specific.

So the test is not *was market research performed* — the file will always say yes. The test is whether it was taken against **this ask**, at **this number** and **this date**; whether the parties answered **separately**, none knowing what the others said; and whether refusal was **free**.

Several fluent parties answering the same question separately let the owner read convergence rather than assess a claim they have no way to test. The owner is a visitor on the contractor's ground and cannot independently test a claim made there. They do not have to. Parties conferring produce one answer with several signatures, and the visitor is exactly as neutral as before. Where answers diverge on an item, that is a tell, not proof: the requirement may read two ways, or the market may be unequal at it. The tell goes back to the document, never to the teams.

The sequence has three acts. The owner affirms first, and blind — the draft is that affirmation, and in the owner's mind it is clear and achievable because all their work has led them to believe so. That belief has no point until the market answers, and it is not completely free: it is priced in calendar, against a period of availability that lapses whether or not anything is built. Next, the market **reads and responds** — both acts are required, because a silent reading with no response returns the owner exactly what they had. Next the owner closes: the final RFP is the owner's affirmation made common, having survived the reading. What was affirmed is what binds.

Duration is not the test. A reading can take an afternoon or eight months. What makes it a reading is that the market is present, the answers are separate, and refusal to affirm is free. A months-long collaboration with a single selected party and an exit path meets none of those conditions, however diligent it looks — and every month spent reaching a number is a month of stop-authority spent. A termination right does not recover it. Dollars are a stock; schedule is not.

The fastest discharge available

The reading looks like added schedule. It is the opposite: it is the cheapest hour at which the owner can be wrong, and the fastest route by which the contracting officer discharges reachability and established outcome certainty.

Being wrong at the draft costs a revision cycle. Being wrong after selection costs the ramp — and by then the calendar spent reaching a number is the same calendar that would have let the owner decline it. Same error, two prices, and the second is

unrecoverable. Every project discovers everything. The only variable is when, and therefore who can still refuse.

- Known at the reading: a requirement, priced, with a free refusal available.
- Discovered after selection: priced by a party facing no competition.
- Discovered at delivery: paid by whoever holds the asset.

After the reading, an unreached floor is the contractor's failure and not the ask's. Before it, the contracting officer is holding an assertion. What the reading produces is a record — several qualified parties, separately, before any of them had an interest in the answer, affirming that the ask read and the floor was there at the number. That record is what a warrant is defended with.

Best Value

Most Ranked Scope Proposed ÷ Fixed Price = Best Value

Best against the offers at the table, at a denominator held constant.

The regulation defines best value as the expected outcome that, in the Government's estimation, provides the greatest overall benefit in response to the requirements. Under the tradeoff approach the Government cannot determine in advance which offer will represent it; only after evaluation may the source selection authority exercise independent judgment, and the rationale need not be quantified. That is value with no common measure — read after proposals are in view, by one party, unquantified. Every protest lives in that gap.

At the other end of the continuum, lowest price technically acceptable gives no credit for exceeding the standard, in turn, the Government realizes no added benefit made available because there is not mechanism for picking it up: unrealized potential written into a source selection method.

The ranked queue is the evaluation factor structure, at item resolution. Every ranked result is a factor. Its measure is the threshold. Its position in the order is its relative importance. All made clear to the proposers and declared before receipt of proposals.

This is not one factor replacing many. It is more factor-level evaluation than a tradeoff with five weighted criteria, not less. What is absent is *weighting after the fact* — the owner encoded relative importance when they set the ranking. Price is evaluated: it is fixed and identical across offers. Quality is evaluated at every line, because the measure attached to

each ranked item is that item's quality standard — a threshold a stranger can read, not an adjectival rating.

With the number fixed, price has nothing left to compete on. The only remaining variable is depth, read mechanically against an order published before anyone proposed. Where the order runs out, an occurrence made unlikely by an extended reach to optional if-possible scope items, a coin flip is more defensible than a discriminator invented after the proposals arrived.

Fixed price does not remove value competition. It is the only condition under which value competition can be read.

Warranted Outcome

The warranty follows the act of deciding what the thing must be.

An owner who publishes what the finished thing must **do** has decided nothing about what it must **be**. The contractor decides that, and in deciding it warrants that the thing does what was published by measure. An owner who writes a direction for *two elevators* has decided being, and now warrants that two elevators will suffice — though nobody wrote that warranty and nobody intended it. It attached the moment the sentence was written.

So the crossing is not a loss of leverage. It is the transfer of a warranty, per line, by operation of the decision. An RFP's warranty allocation is fully determined by its own text, and neither party's intent enters.

The measure is what is warranted. Not the result in the abstract, not what a reasonable person would consider adequate — the stated procedure, conditions, instrument, and threshold. Which is why the warranty is finite and releasable: the contractor knows exactly what they warranted, because they can read it, and so can a stranger with access to the measure. And an unmeasured outcome is warranted by nobody. Not weakly — not at all. The sweeping quality language with no measure attached is the line where the owner got no protection, not the most.

The warranty attaches per line. A punch list item is a failed reading. The owner does not argue workmanship — an adjective with no procedure, no instrument, and no threshold, resolvable only by someone's judgment, which is will, which is what the measure was there to replace. The owner points at the measure and says it was not met. A competent contractor already knows and why — it lies in the measure. There is no third position.

This is also what makes the transaction behave like a product purchase rather than a service. A service is steered as it is produced, decisions as to measure come late and are discovered over time, so steering is what change orders are for. A product is defined before it is made, and the buyer's remaining acts are receipt and verification. Where the owner steers, they decide *being* in small increments — each one agreed rather than imposed, each one taking back a share of the warranty the owner thought they had, none of them recorded as a loss.

What the diagnostic returns

Findings with addresses. Not a score.

A missing measure names the line. An unread number names the IGE. A reading taken after competition opened names the hour. Every void has a holder, and naming them is the whole of the first law.

A solicitation with more elements present is not thereby better, and no owner is required to close every exposure — one may be knowingly carried. What cannot happen is holding exposure nobody knows they are holding.

Where this has been done

The most publicly recorded by the Government: The Department of Energy procured the Research Support Facility at NREL (NLR) with a fixed price set before the performance goals were established, a scope ranked in three tiers — mission critical, highly desirable, if possible — and a draft RFP given to the final competitors before competition, because the agency did not know whether it had hit the mark. Performance goals rather than technical specifications, with substantiation criteria attached to the goals. The owner's stated reason for being able to use a firm fixed-price contract was that the contractors, having the measures, knew exactly what was wanted, which reduced their risk.

Five of the six elements are on the federal record for that project. The sixth — a costless no — is implied by the structure but not stated in the publications.

- *The Design-Build Process for the Research Support Facility*, NREL — https://docs.nrel.gov/docs/fy12osti/51387.pdf
- National Research Council, *Achieving High-Performance Federal Facilities*, 2011, Appendix G — https://www.nationalacademies.org/read/13140/chapter/16

The RSF was procured through a method known as the 3PQ Acquisition & Management System©. The method rests on the physics, not the other way around.

How to argue with this

No obligation. Nothing here can be obeyed or violated. The laws do not demand recognition any more than gravity does. These statements describe how promises behave and predict what a structure will cost the party holding the asset.

Falsifiable. Each law could be falsified by a project. If you have that project, bring it into this discussion with these tests — where it can be seen.

Already revised once. I built and sold education under a delivery-model banner I now believe disperses the first law. The structure asserted itself against my own commercial position, so I moved.

Who owns the consequence · when does it close · who can hold the line.

Appendix

Crosswalk: RFO provisions · the six elements · the three laws

Current as of August 26, 2026. The RFO is being implemented through rolling class deviations, agency by agency, ahead of formal rulemaking. Section numbers move. Verify against your agency's current deviation before relying on any citation here.

How to read the third column. This is not an alignment exercise. Each provision is read for what it closes and what it leaves open. Several of the strongest RFO provisions close nothing on their own — they create the condition under which something can be closed, and leave the closing to the project owner.

Part 16 · Contract types

What it requires. Fixed-price with performance-based considerations is the default and preferred contract type (RFO 16.102), implementing EO 14402 (April 30, 2026). Anything else above the agency threshold requires written justification under 16.103 and agency-head approval, delegable only to the chief acquisition officer or a non-career SES. DoD implements through Class Deviation 2026-O0045, Rev. 1. Agencies are directed to consider fixed pricing for portions of a requirement even where the whole cannot be structured that way.

What the physics says. This closes **element 1** — and nothing else.

A fixed price is one dimension closed. Scope is the dimension with no native unit, and a closed number against an open scope is the rate law's engine: every scope question is a dollars question, every dollars answer reopens scope, and the exchange ends when the calendar is exhausted rather than when the answer arrives. The mandate therefore produces one of two things depending on what it is attached to. Attached to a fixed *solution*, competition can only run downward on price, and the reach never surfaces. Attached to an unmeasured *ask*, the scope reopens after award, priced by a party facing no competition.

The regulation cannot supply what makes fixed price work. **A common measure per result is authored in the owner's domain and nowhere else.**

Worth noting for the recurrence: the fixed-price preference dates to FASA in 1994 and performance-based contracting to the FY2001 NDAA. Thirty-two years of mandating the outcome without the conditions that produce it. Under Law II, that is what a restatement of an expectation looks like when the mechanism is unnamed.

Part 7 · Acquisition planning & market research

What it requires and permits. Under FAR Case 2026-002 (91 FR 37636, June 23, 2026), market research moves out of Part 10 into proposed subpart 7.2, and the industry-engagement requirements are relocated from FAR 15.201 to FAR 7.105. The market research clause moves from 52.210-1 to 52.207-7. Proposed FAR 7.201, Market research requirements, carries the obligation: agencies must describe their legitimate needs and conduct research appropriate to the circumstances before developing requirements documents and before soliciting. Paragraph (c) receives text relocated from FAR 1.102-2(a)(4), implementing section 887 of the FY2016 NDAA (41 U.S.C. 1703 note, *Effective Communication Between Government and Industry*) — agencies should engage in responsible and constructive exchanges with industry. The Council removed the enumerated techniques and considerations entirely: the method is the team's choice, and the stated reason is to free agencies to implement the best approach rather than follow a prescriptive process regardless of the marketplace.

What the physics says. This is the authority for **element 5** — and the relocation strengthens it. Planning, market research and industry engagement now sit in one part, which is where the ask is formed. The exchanges that used to live in the source selection part are no longer there at all.

Every condition the reading requires is permitted. Separate engagement: the method is the team's choice, and nothing requires meeting every possible offeror or disclosing one party's answers to another. Improving the requirement on what industry reports: that is the stated purpose of the statutory communication mandate. And the authority runs until competition begins, which is precisely where the instrument stops working — before that point refusal is free, so a yes is evidence; after selection there is nothing left to decline and every affirmation is compliance.

What it does not supply. The regulation authorizes the research, and having removed the prescribed techniques, it now says less than ever about what to ask. A file that reports capability in general, against no specific ask and no specific number, discharges nothing. The compound question is what converts the authority into a reading:

Is the RFP clear, and if so, can the required scope be achieved at the price (IGE) and time stated?

The diagnostic test on this part is not *was market research performed*. It is: taken against **this ask**, at **this number** and **this date**; answers given **separately**; refusal **free**.

Part 11 · Describing agency needs

What it requires. Describe requirements in terms that permit full and open competition, and state them in terms of function, performance, or essential physical characteristics rather than by specifying a solution. The RFO eliminated large portions of the prior part, including the delivery-and-performance-schedules subpart and the variation-in-quantity subpart, and streamlined the market acceptance criteria.

What the physics says. This is where **element 4** and **element 6** live, and where the regulation's language is most often satisfied in form and missed in substance.

Performance means achievement of a stated measure. It is not subjective, does not mean effort, diligence, or conduct. A performance requirement without a procedure, an instrument, and a threshold is not a loose measure; it is the absence of measure because it lacks objectivity, it only wears the name. Under Law II a promise binds only to the extent it was made, so an unmeasured requirement binds nobody and is satisfied by the ordinary.

And the warranty follows the act of deciding what the thing must be. An owner who publishes what the finished thing must *do* has decided nothing about what it must *be*; the contractor decides that and warrants the doing. An owner who writes *two elevators* has decided *being*, and now warrants that two elevators move the people. Every line written as a means rather than as a result moves the warranty back to the government, per line, by operation of the text — regardless of what the solicitation is called.

The diagnostic test on this part is line by line: does this line have a procedure, conditions, an instrument, a threshold, and a knowable release? If not, it is a defect with a known repair, curable at the desk, in the government's own domain, at no cost to anyone, before the competition begins.

Part 15 · Source selection

Pre-solicitation exchanges with industry are no longer in this part — FAR 15.201 moves to FAR 7.105. Engagement belongs to planning; evaluation belongs to selection.

What it requires. Best value is the expected outcome that, in the Government's estimation, provides the greatest overall benefit in response to the requirement (RFO 2.101). The continuum runs from lowest price technically acceptable, with guardrails (15.103-2), through the tradeoff approach (15.103-1), to highest technically rated with fair and reasonable price (15.103-3). Every source selection must evaluate cost or price and quality of product or service (15.104(b)(1)). The solicitation must state the relative importance of all factors and significant subfactors and their relationship to cost or price (15.104(d)).

Under tradeoff, the source selection decision is the authority's independent judgment on a comparative assessment, documented with rationale — and the documentation need not quantify the tradeoffs.

What the physics says. This is where the absence of a common measure is written into the regulation's own account of itself.

In the Government's estimation is an assertion on one party's side of the table. Under tradeoff, best value cannot be determined in advance; it is read after proposals are in view, by one party, with a rationale that need not be quantified. That is value with no common measure, and it is where protests live. At the other end, LPTA gives no credit for exceeding the standard — unrealized potential written into a selection method.

Element 3 answers 15.104(d) at item resolution. The ranked scope *is* the evaluation factor structure: every ranked result is a factor, its measure is the threshold, its position in the order is its relative importance, all declared before receipt of proposals. That is more factor-level evaluation than a tradeoff with five weighted criteria, not less. What is absent is weighting after the fact.

Both statutory factors are evaluated. Price: fixed and identical across offers. Quality: at every line, because the measure attached to each ranked item is that item's quality standard — a threshold a stranger can read, not an adjectival rating.

Most Ranked Scope Proposed ÷ Fixed Price = Best Value — best against the offers at the table, at a denominator held constant. The award rationale is then a published order applied mechanically, rather than a judgment exercised after the proposals arrived.

Element 3 also has a life span. The ranking governs from publication through competition and flattens at award: everything committed in the proposal becomes one list and all are converted to contractually required, so a volunteered *if-possible* looses its tier designation and is as enforceable as a all of the *required* tier. A ranked queue that survived into the contract would make *if-possible* a standing excuse.

Summary

Element	Where the authority already exists	What the regulation does not supply
1 · Fixed price	RFO 16.102, EO 14402	What the number is fixed *against*
2 · Fixed date	Period of availability; solicitation terms	An enforcer; time generates no refusal on its own
3 · Ranked scope, flattened at award	RFO 15.104(d) relative importance	The ranking itself — the owner's exclusive act
4 · Common measure and its discharge	Part 11 function/ performance description	The procedure, instrument, threshold, and release
5 · Costless no	Subpart 7.2, until competition begins	The question to ask
6 · Exclusive domains	Part 11 solution-neutral description	Restraint after award — Law III

The pattern. The regulation supplies authority and mandates outcomes. It does not, and cannot, supply the measure because it is project dependent. Every element the RFO leaves open is authored, for the project, in the owner's domain, before the solicitation issues, in the one window where the whole ask is held at uniform clarity, and leverages competition.

What the crosswalk does not claim. That the RFO was written with this in mind, or that following it produces these elements. The provisions permit the reading; nothing requires it. A file can be fully compliant with Parts 10, 11, 15 and 16 and close none of the six elements.

Who owns the consequence · when does it close · who can hold the line.